

Math 110 HW #1

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Axler 1B: 5. Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbf{R}$. Define an addition and scalar multiplication on $\mathbf{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbf{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbf{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbf{R}$? Explain.

Solution:

By definition, for any three elements $u, v, w \in V$, it must satisfy the associativity of addition (one of the 8 axioms):

$$(u + v) + w = u + (v + w)$$

Let us choose the following three elements from our set V :

- $u = \infty$
- $v = -\infty$
- $w = 1$ (since $1 \in \mathbf{R}$, it is also in $\mathbf{R} \cup \{\infty, -\infty\}$)

LHS: $(u + v) + w = (\infty + (-\infty)) + 1$. According to the given rule $\infty + (-\infty) = 0$, we substitute 0 into the parentheses:

$$= 0 + 1 = 1$$

RHS: $u + (v + w) = \infty + ((-\infty) + 1)$. According to the given rule $(-\infty) + t = -\infty$ for any real number $t \in \mathbf{R}$, we substitute $-\infty$ for the expression inside the parentheses:

$$= \infty + (-\infty)$$

Using the given rule $\infty + (-\infty) = 0$, this evaluates to:

$$= 0$$

Since $\text{LHS} = 1 \neq \text{RHS} = 0$, we have shown that:

$$(\infty + (-\infty)) + 1 \neq \infty + ((-\infty) + 1)$$

Because the associativity of addition property fails to hold for all elements in the set, $\mathbf{R} \cup \{\infty, -\infty\}$ is not a vector space over $\mathbf{R}$.

Axler 1C: 6. Give an example of a nonempty subset U of $\mathbf{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbf{R}^2$.

Solution:

To find a subset U that is closed under scalar multiplication but is **not** a subspace, we must find a set where scalar multiplication always stays inside the set, but **addition fails** (it is not closed under addition).

U can be the union of all the points on the x -axis and the y -axis in the Cartesian plane:

$$U = \{(x, y) \in \mathbf{R}^2 : x = 0 \text{ or } y = 0\}$$

Nonempty: This subset is nonempty because it contains elements like $(0, 0)$, $(1, 0)$, and $(0, 1)$.

Closure under Scalar Multiplication: Let us choose an arbitrary vector $v \in U$ and an arbitrary scalar $\lambda \in \mathbf{R}$.

$$\lambda v = \lambda(x, y) = (\lambda x, \lambda y)$$

- If $x = 0$, then $\lambda x = \lambda(0) = 0$. Thus, $(\lambda x, \lambda y) = (0, \lambda y)$, which satisfies the condition of being in U since x is still 0.
- If $y = 0$, then $\lambda y = \lambda(0) = 0$. Thus, $(\lambda x, \lambda y) = (\lambda x, 0)$, which satisfies the condition of being in U since y is still 0.

In both cases, $\lambda v \in U$. Therefore, U is closed under scalar multiplication.

Closure under Vector Addition: We can show a counterexample. Let us choose:

- $u = (1, 0)$ (This is in U because the y -coordinate is 0)
- $v = (0, 1)$ (This is in U because the x -coordinate is 0)

$$u + v = (1, 0) + (0, 1) = (1 + 0, 0 + 1) = (1, 1)$$

Since the resulting vector $(1, 1)$ does not have 0 as either of its coordinates, $(1, 1) \notin U$. Therefore, U is not closed under vector addition.

Since $U = \{(x, y) \in \mathbf{R}^2 : x = 0 \text{ or } y = 0\}$ is a nonempty subset of $\mathbf{R}^2$ that is closed under scalar multiplication but not closed under vector addition, it is **not a subspace** of $\mathbf{R}^2$.

Axler 2A: 2. Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

Solution:

We are given that the list (v_1, v_2, v_3, v_4) spans V . This means that any vector in V can be built out of linear combinations of v_1, v_2, v_3 , and v_4 .

To show that the new list $(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$ also spans V , we only need to show that each of the original building blocks (v_1, v_2, v_3, v_4) can themselves be built (written as a linear combination) using the elements of the new list.

Let us look at each original vector one by one and try to write it as a combination of the vectors in the new list: $w_1 = v_1 - v_2$, $w_2 = v_2 - v_3$, $w_3 = v_3 - v_4$, and $w_4 = v_4$.

$$v_4 = 1(w_4)$$

$$v_3 = 1(w_3) + 1(w_4) = v_3 - v_4 + v_4$$

$$v_2 = 1(w_2) + 1(w_3) + 1(w_4) = v_2 - v_3 + v_3 - v_4 + v_4$$

$$v_1 = 1(w_1) + 1(w_2) + 1(w_3) + 1(w_4) = v_1 - v_2 + v_2 - v_3 + v_3 - v_4 + v_4$$

Since the original list (v_1, v_2, v_3, v_4) spans V , there exists some scalars $a_1, a_2, a_3, a_4 \in \mathbf{F}$ such that arbitrary vector $v \in V$ can be written as:

$$v = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$$

Now, substitute our expressions for v_1, v_2, v_3 , and v_4 in terms of w_1, w_2, w_3, w_4 into this equation:

$$v = a_1(w_1 + w_2 + w_3 + w_4) + a_2(w_2 + w_3 + w_4) + a_3(w_3 + w_4) + a_4(w_4)$$

Since this is a vector space, we can use the distributive and associative properties to collect the coefficients for each w_i vector:

$$\begin{aligned} v &= a_1w_1 + a_1w_2 + a_1w_3 + a_1w_4 + a_2w_2 + a_2w_3 + a_2w_4 + a_3w_3 + a_3w_4 + a_4w_4 \\ v &= a_1w_1 + (a_1 + a_2)w_2 + (a_1 + a_2 + a_3)w_3 + (a_1 + a_2 + a_3 + a_4)w_4 \end{aligned}$$

Since $a_1, a_2, a_3, a_4 \in \mathbf{F}$, and fields are closed by addition, their sums are also scalars within the field $\mathbf{F}$:

- $x_2 = a_1 \in \mathbf{F}$
- $c_2 = a_1 + a_2 \in \mathbf{F}$
- $c_3 = a_1 + a_2 + a_3 \in \mathbf{F}$
- $c_4 = a_1 + a_2 + a_3 + a_4 \in \mathbf{F}$

Substituting these new scalar constants back into our equation gives:

$$v = c_1w_1 + c_2w_2 + c_3w_3 + c_4w_4$$

Replacing w_1, w_2, w_3, w_4 back with their original definitions shows that:

$$v = c_1(v_1 - v_2) + c_2(v_2 - v_3) + c_3(v_3 - v_4) + c_4v_4$$

Because every arbitrary vector $v \in V$ can be written as a linear combination of the vectors in the list $(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$, we conclude that **this list also spans V** .

Axler 2A: 14. Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that the list $v_1, \dots, v_m$ is linearly independent if and only if the list $w_1, \dots, w_m$ is linearly independent.

Solution:

Definition: Any list of vectors $(u_1, \dots, u_m)$ is **linearly independent** if the only choice of scalars $c_1, \dots, c_m \in \mathbf{F}$ that satisfies the vector equation:

$$c_1u_1 + \dots + c_mu_m = 0$$

is the trivial solution where every scalar is exactly zero:

$$c_1 = 0, c_2 = 0, \dots, c_m = 0$$

Setup: These are the explicit definitions of each vector w_k given in the problem:

$$\begin{aligned} w_1 &= v_1 \\ w_2 &= v_1 + v_2 \\ w_3 &= v_1 + v_2 + v_3 \\ &\vdots \\ w_m &= v_1 + v_2 + \cdots + v_m \end{aligned}$$

We can also write each original vector v_k in terms of the new vectors w_k by subtraction:

$$\begin{aligned} v_1 &= w_1 \\ v_2 &= w_2 - w_1 \\ v_3 &= w_3 - w_2 \\ &\vdots \\ v_k &= w_k - w_{k-1} \quad \text{for } k \geq 2 \end{aligned}$$

($\Rightarrow$) Assume $v_1, \dots, v_m$ is linearly independent, prove that $w_1, \dots, w_m$ is linearly independent.

$$c_1 w_1 + c_2 w_2 + c_3 w_3 + \cdots + c_m w_m = 0$$

$$c_1(v_1) + c_2(v_1 + v_2) + c_3(v_1 + v_2 + v_3) + \cdots + c_m(v_1 + v_2 + \cdots + v_m) = 0$$

$$(c_1 + c_2 + c_3 + \cdots + c_m)v_1 + (c_2 + c_3 + \cdots + c_m)v_2 + \cdots + (c_{m-1} + c_m)v_{m-1} + c_m v_m = 0$$

Since we assumed that the list $v_1, \dots, v_m$ is linearly independent, the coefficients of each v_k term must be 0, meaning $c_m = 0$. Then, we can work our way backwards:

$$c_{m-1} + c_m = 0 \implies c_{m-1} + 0 = 0 \implies c_{m-1} = 0$$

$$\vdots$$

$$c_1 + c_2 + \cdots + c_m = 0 \implies c_1 + 0 + \cdots + 0 = 0 \implies c_1 = 0$$

Because every single coefficient c_k must equal 0, the list $w_1, \dots, w_m$ is linearly independent.

($\Leftarrow$) Assume $w_1, \dots, w_m$ is linearly independent, prove that $v_1, \dots, v_m$ is linearly independent.

$$d_1 v_1 + d_2 v_2 + d_3 v_3 + \cdots + d_m v_m = 0$$

$$d_1(w_1) + d_2(w_2 - w_1) + d_3(w_3 - w_2) + \cdots + d_m(w_m - w_{m-1}) = 0$$

$$(d_1 - d_2)w_1 + (d_2 - d_3)w_2 + \cdots + (d_{m-1} - d_m)w_{m-1} + d_m w_m = 0$$

Since we assumed that the list $w_1, \dots, w_m$ is linearly independent, the coefficients of each w_k term must be 0, meaning $d_m = 0$. Then, we can work our way backwards:

$$\begin{aligned} d_{m-1} - d_m = 0 &\implies d_{m-1} - 0 = 0 \implies d_{m-1} = 0 \\ &\vdots \\ d_1 - d_2 = 0 &\implies d_1 - 0 = 0 \implies d_1 = 0 \end{aligned}$$

Because every single coefficient d_k must equal 0, the list $v_1, \dots, v_m$ is linearly independent.

Since both direction is true, the list $v_1, \dots, v_m$ is linearly independent **if and only if** the list $w_1, \dots, w_m$ is linearly independent.

Axler 2B: 3. (a) Let U be the subspace of $\mathbf{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of U .

(b) Extend the basis in (a) to a basis of $\mathbf{R}^5$.

(c) Find a subspace W of $\mathbf{R}^5$ such that $\mathbf{R}^5 = U \oplus W$.

Solution:

Definitions:

- **Basis:** A basis of a vector space V (or subspace) is a list of vectors that is linearly independent and spans V . Alternatively, if V is finite-dimensional with a known dimension n , any linearly independent list of exactly n vectors automatically spans V and forms a basis.
- **Direct Sum ($\oplus$):** A vector space V is the direct sum of two subspaces U and W (written $V = U \oplus W$) if every vector $v \in V$ can be written uniquely as a sum $u + w$, where $u \in U$ and $w \in W$. This happens if and only if $U + W = V$ and $U \cap W = \{0\}$.

(a) We can rewrite $(x_1, x_2, x_3, x_4, x_5) \in U$ by substituting in $x_1 = 3x_2$ and $x_3 = 7x_4$ as

$$(3x_2, x_2, 7x_4, x_4, x_5) = x_2(3, 1, 0, 0, 0) + x_4(0, 0, 7, 1, 0) + x_5(0, 0, 0, 0, 1)$$

This means x_2, x_4 , and x_5 are free variables, and every single vector in U is a linear combination of these three vectors:

$$u_1 = (3, 1, 0, 0, 0), \quad u_2 = (0, 0, 7, 1, 0), \quad u_3 = (0, 0, 0, 0, 1)$$

Since these vectors span U and are linearly independent, the basis of U is:

$$\left((3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1) \right)$$

(b) The standard basis of $\mathbf{R}^5$ consists of the five standard unit vectors:

$$e_1 = (1, 0, 0, 0, 0), \quad e_2 = (0, 1, 0, 0, 0), \quad e_3 = (0, 0, 1, 0, 0), \quad e_4 = (0, 0, 0, 1, 0), \quad e_5 = (0, 0, 0, 0, 1)$$

Since $\dim(\mathbf{R}^5) = 5$ and our basis of U only has 3 vectors, we need to add 2 more vectors from the standard basis that do not already lie in the span of U . We can try $e_1 = (1, 0, 0, 0, 0)$ and $e_3 = (0, 0, 1, 0, 0)$ and see if the new list of 5 vectors is linearly independent:

$$x_2(3, 1, 0, 0, 0) + x_4(0, 0, 7, 1, 0) + x_5(0, 0, 0, 0, 1) + x_1(1, 0, 0, 0, 0) + x_3(0, 0, 1, 0, 0) = (0, 0, 0, 0, 0)$$

$$(3x_2 + x_1, x_2, 7x_4 + x_3, x_4, x_5) = (0, 0, 0, 0, 0)$$

We get $x_2, x_4, x_5 = 0$, therefore, $x_1, x_3 = 0$. Since all coefficients is equal to 0, this extended list of 5 vectors is linearly independent. Any linearly independent list of length 5 in a 5-dimensional space automatically forms a basis. So, the extended basis of $\mathbf{R}^5$ is:

$$\left((3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1), (1, 0, 0, 0, 0), (0, 0, 1, 0, 0) \right)$$

(c) Theorem 2.33 states that suppose V is finite-dimensional and U is a subspace of V , then there is a subspace W of V such that $V = U \oplus W$.

- i) Start with a basis $(u_1, \dots, u_m)$ of the subspace U .
- ii) Extend this list to a full basis $(u_1, \dots, u_m, w_1, \dots, w_n)$ of the entire space V .
- iii) Define the new subspace W as the span of the appended vectors:
 $W = \text{span}(w_1, \dots, w_n)$.

If we construct W this way, the theorem guarantees that every vector in V can be uniquely written as a sum of a vector from U and a vector from W , satisfying the definition of a direct sum ($V = U \oplus W$). Here, we can define our subspace W to be the span of these two added vectors from part (b):

$$\begin{aligned} W &= \text{span}\left((1, 0, 0, 0, 0), (0, 0, 1, 0, 0)\right) \\ &= \{(x_1, 0, x_3, 0, 0) \in \mathbf{R}^5 : x_1, x_3 \in \mathbf{R}\} \end{aligned}$$

This guarantees $\mathbf{R}^5 = U \oplus W$ because our full 5-vector list from part (b) is a basis of $\mathbf{R}^5$, any vector $v \in \mathbf{R}^5$ can be written uniquely as:

$$v = \underbrace{c_1u_1 + c_2u_2 + c_3u_3}_{\in U} + \underbrace{c_4w_1 + c_5w_2}_{\in W}$$

Since the first three terms combine to form a single unique vector in U and the last two terms combine to form a single unique vector in W , every vector in $\mathbf{R}^5$ can be expressed uniquely as $u + w$ where $u \in U$ and $w \in W$. This satisfies the definition of a direct sum. Therefore, the subspace $W = \{(x_1, 0, x_3, 0, 0) \in \mathbf{R}^5 : x_1, x_3 \in \mathbf{R}\}$ satisfies $\mathbf{R}^5 = U \oplus W$.

Axler 2B: 7. Suppose v_1, v_2, v_3, v_4 is a basis of V . Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V .

Solution:

We are given that (v_1, v_2, v_3, v_4) is a basis of V . This tells us that:

- The dimension of V is exactly 4 ($\dim V = 4$) because the length of its basis is 4.
- The list (v_1, v_2, v_3, v_4) is linearly independent.

Our new list $(v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4)$ also contains exactly 4 vectors. Because the length of our new list matches the dimension of the space ($\dim V = 4$), we can show that the new list is linearly independent (which automatically implies it spans because its length is 4).

$$w_1 = v_1 + v_2, \quad w_2 = v_2 + v_3, \quad w_3 = v_3 + v_4, \quad w_4 = v_4$$

To test if the list (w_1, w_2, w_3, w_4) is linearly independent:

$$\begin{aligned} c_1 w_1 + c_2 w_2 + c_3 w_3 + c_4 w_4 &= 0 \\ c_1(v_1 + v_2) + c_2(v_2 + v_3) + c_3(v_3 + v_4) + c_4(v_4) &= 0 \\ c_1 v_1 + c_1 v_2 + c_2 v_2 + c_2 v_3 + c_3 v_3 + c_3 v_4 + c_4 v_4 &= 0 \\ (c_1)v_1 + (c_1 + c_2)v_2 + (c_2 + c_3)v_3 + (c_3 + c_4)v_4 &= 0 \end{aligned}$$

Since the list (v_1, v_2, v_3, v_4) is a basis, therefore automatically linearly independent, the coefficients of each v_k term must be 0. Then, we can work our way backwards:

$$\begin{aligned} c_1 &= 0 \\ c_1 + c_2 = 0 &\implies c_2 = 0 \\ c_2 + c_3 = 0 &\implies c_3 = 0 \\ c_3 + c_4 = 0 &\implies c_4 = 0 \end{aligned}$$

Since $c_1 = c_2 = c_3 = c_4 = 0$ if $c_1 w_1 + c_2 w_2 + c_3 w_3 + c_4 w_4 = 0$, the list $(v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4)$ is linearly independent.

By the dimension theorem, any linearly independent list whose length equals the dimension of the vector space automatically spans the space and forms a basis. Therefore, this list **is also a basis of V** .

Axler 3A: 8. Give an example of a function $\varphi : \mathbf{R}^2 \rightarrow \mathbf{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbf{R}$ and all $v \in \mathbf{R}^2$ but φ is not linear.

This exercise and the next exercise show that neither homogeneity nor additivity alone is enough to imply that a function is a linear map.

Solution:

Definitions:

- **Linear Map:** A function $\varphi : V \rightarrow W$ is a **linear map** (or transformation) if for all vectors $u, v \in V$ and all scalars $a \in \mathbf{F}$:

(a) **Additivity:** $\varphi(u + v) = \varphi(u) + \varphi(v)$

(b) **Homogeneity:** $\varphi(av) = a\varphi(v)$

Setup: We need a function $\varphi(x, y)$ that takes a pairs of numbers and outputs a single number. It must:

(a) Scalar multiplication: $\varphi(ax, ay) = a\varphi(x, y)$ (Homogeneity)

(b) Fail to vector addition: $\varphi(u + v) \neq \varphi(u) + \varphi(v)$ (Non-additivity)

We try the simplest function we can think of and test:

$$\varphi(x, y) = x + y$$

$$\varphi(ax, ay) = ax + ay = a(x + y) = a\varphi(x, y)$$

$$\varphi(u + v) = (x_1 + x_2) + (y_1 + y_2) = (x_1 + y_1) + (x_2 + y_2) = \varphi(u) + \varphi(v)$$

This satisfies additivity, but we need additivity to break. Let us try with exponents, because we know from algebra that $(x_1 + x_2)^n \neq x_1^n + x_2^n$.

$$\varphi(x, y) = x^2 + y^2$$

$$\varphi(ax, ay) = (ax)^2 + (ay)^2 = a^2x^2 + a^2y^2 = a^2(x^2 + y^2) = a^2\varphi(x, y)$$

$$\varphi(u + v) = (x_1 + x_2)^2 + (y_1 + y_2)^2 = x_1^2 + 2x_1x_2 + x_2^2 + y_1^2 + 2y_1y_2 + y_2^2 \neq \varphi(u) + \varphi(v)$$

This already violates our requirement of homogeneity. So, we see that:

- If the variables inside have a power of 1 (like $x + y$), the function scales by a^1 but remains additive.
- If the variables inside have a higher power (like $x^2 + y^2$), it breaks additivity, but it scales by a^2 , which ruins homogeneity.

We need something that uses higher powers inside to break additivity, but cancels out those powers globally so that it still scales by exactly a^1 . Let's try a square root to cancel the squares out: $\sqrt{x^2 + y^2}$. But for negative numbers like $a = -1$:

$$\varphi(-1x, -1y) = \sqrt{(-1x)^2 + (-1y)^2} = \sqrt{x^2 + y^2} = \varphi(x, y)$$

It is not equal to $-1\varphi(x, y)$ because negative signs get eaten by the square roots, which would fail homogeneity for negative scalars. So, let us try an odd power and an odd root, because odd roots preserve negative signs ($\sqrt[3]{-8} = -2$).

Let us define a function $\varphi : \mathbf{R}^2 \rightarrow \mathbf{R}$ using cubes and a cube root:

$$\varphi(x, y) = \sqrt[3]{x^3 + y^3}$$

By using cubes inside, we guarantee addition will fail ($1^3 + 1^3 \neq (1+1)^3$).
By using the cube root, it will preserve negative signs.

Homogeneity Let $a \in \mathbf{R}$ be any scalar, and let $v = (x, y) \in \mathbf{R}^2$ be any vector.

$$\begin{aligned}\varphi(av) &= \varphi(ax, ay) = \sqrt[3]{(ax)^3 + (ay)^3} \\ &= \sqrt[3]{a^3x^3 + a^3y^3} \\ &= \sqrt[3]{a^3(x^3 + y^3)} \\ &= \sqrt[3]{a^3} \cdot \sqrt[3]{x^3 + y^3} \\ &= a \sqrt[3]{x^3 + y^3} \\ &= a\varphi(v)\end{aligned}$$

Additivity Fail Let $u = (1, 0)$ and $v = (0, 1)$ from $\mathbf{R}^2$.

$$T(u)+T(v) = \varphi(1, 0)+\varphi(0, 1) = \sqrt[3]{1^3 + 0^3} + \sqrt[3]{0^3 + 1^3} = \sqrt[3]{1} + \sqrt[3]{1} = 1+1 = 2$$

$$T(u+v) = \varphi(1, 1) = \sqrt[3]{1^3 + 1^3} = \sqrt[3]{1+1} = \sqrt[3]{2}$$

Since $2 \neq \sqrt[3]{2}$, we have shown that $\varphi(u+v) \neq \varphi(u) + \varphi(v)$. Because additivity fails, the function φ is not a linear map.

Axler 3B: 10. Suppose $v_1, \dots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans range T .

Solution:

Definitions:

- **Spans:** To say a list $(v_1, \dots, v_n)$ spans V means that *every* vector $v \in V$ can be written as a linear combination of those elements:

$$v = c_1v_1 + \dots + c_nv_n \quad \text{for some scalars } c_i \in \mathbf{F}$$

- **Linear Map** ($T \in \mathcal{L}(V, W)$): This tells us that T is a function from V to W that preserves structure. Specifically, it satisfies:

$$T(u+v) = Tu + Tv \quad \text{and} \quad T(cv) = cTv$$

Combining these properties, a linear map distributes perfectly across any linear combination:

$$T(c_1v_1 + \dots + c_nv_n) = c_1Tv_1 + \dots + c_nTv_n$$

- **Range of T ($\text{range } T$):** The range (or image) is the set of all outputs that T can actually hit in the target space W . By definition:

$$\text{range } T = \{w \in W : w = Tv \text{ for some } v \in V\}$$

Setup: Let w be an arbitrary vector in $\text{range } T$. By the definition of range of a linear map, $w \in \text{range } T$, there exist some vector $v \in V$ such that:

$$w = Tv$$

There also exist scalars $c_1, \dots, c_n \in \mathbf{F}$ such that:

$$v = c_1v_1 + \dots + c_nv_n$$

Now, substitute this expression for v back into the first equation for w :

$$w = T(c_1v_1 + \dots + c_nv_n)$$

Because T is a linear map ($T \in \mathcal{L}(V, W)$), we can apply the property of linearity to distribute T across the sum and pull out each scalar multiplier:

$$w = c_1Tv_1 + \dots + c_nTv_n$$

By definition of a span, this means that the list $Tv_1, \dots, Tv_n$ spans **range** T .

Axler 3D: 11. Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that

$$ST \text{ is invertible} \iff S \text{ and } T \text{ are invertible.}$$

Solution:

Definitions:

- $A \in \mathcal{L}(V)$ is **invertible** if there exists an operator $A^{-1} \in \mathcal{L}(V)$ such that $AA^{-1} = I$ and $A^{-1}A = I$.
- If V is **finite-dimensional**, we know that A is invertible $\iff A$ is injective (one-to-one) $\iff A$ is surjective (onto). This means we can simply look at the null space (injectivity: $Av = 0 \implies v = 0$).

($\Leftarrow$) Assume S and T are invertible, prove that ST is also invertible.

This means there exists S^{-1} and T^{-1} . Let us guess $(ST)^{-1} = T^{-1}S^{-1}$.

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = S(I)S^{-1} = SS^{-1} = I$$

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}(I)T = T^{-1}T = I$$

($\Rightarrow$) Assume ST is invertible, prove that both S and T are also invertible.

We cannot find an algebraic formula for S^{-1} or T^{-1} directly out of thin air. But because V is finite-dimensional, an operator is invertible if and only if it is injective (its null space is $\{0\}$).

To show T is injective, let us assume there exists some vector $v \in V$ such that:

$$Tv = 0$$

Apply the linear operator S to both sides of this equation:

$$S(Tv) = S(0)$$

Composing functions on the left and applying the zero-property of linear maps on the right gives:

$$(ST)v = 0$$

By our initial assumption, the operator ST is invertible, which means ST is strictly injective. By definition of injectivity, the only vector that can map to 0 under ST is the zero vector itself:

$$v = 0$$

Thus, we have $Tv = 0 \implies v = 0$, proving that T is injective. Since V is finite-dimensional, T is **invertible**, meaning T^{-1} exists. Let's try multiplying:

$$(ST)T^{-1} = S(TT^{-1}) = S(I) = S$$

Because the set of invertible operators is closed under composition, and both ST and T^{-1} are verified invertible operators, their product $(ST)T^{-1}$ must also be an invertible operator. Since $S = (ST)T^{-1}$, we conclude that S is **invertible**.

Therefore, ST is invertible if and only if both S and T are invertible.

Non-Textbook: 10. Consider a field with three elements: $\{0, 1, a\}$. Determine the addition and multiplication tables for this field. For each non-obvious entry, briefly explain which field axiom or property forces your answer.

Solution:

Addition uses 0 as the additive identity element ($0 + x = x$ for any $x \in \mathbf{F}$). This gives us our obvious boundary row and column: $0 + 0 = 0$, $0 + 1 = 1$, $0 + a = a$, $1 + 0 = 1$, and $a + 0 = a$.

Let's find $1 + 1$.

- Can $1 + 1 = 1$? 1 must have an additive inverse, and because adding the additive inverse of 1 to both sides would force $1 = 0$, $1 + 1$ cannot equal 1. **No.**

- Can $1 + 1 = 0$? 1 must have an additive inverse, so 1 must be the additive inverse of itself because 0 could not be due to $0 + 1 = 1 (\neq 0)$. 0 would have been the additive inverse of 1 if $0 + 1 = 0$ (is 0). However, if 1 is its own additive inverse ($1 + 1 = 0$), then let us look at what happens to the entry $1 + a$. In a field's addition table, every element must appear exactly once in each row and column. If $1 + 1 = 0$, then it leaves a as the only option for the final slot in the row for 1, forcing $1 + a = a$. But by subtracting a from both sides, $1 + a = a \implies 1 = 0$, a contradiction! **No**.
- Therefore, $1 + 1$ cannot be 1, and it cannot be 0. It is structurally forced to be $1 + 1 = a$.

Let's find $1 + a$ and $a + 1$:

- Now look at the row for 1. We have filled $1 + 0 = 1$ and $1 + 1 = a$. The entry $1 + a$ must be the last remaining element of our field to satisfy the unique row property. Thus, $1 + a = 0$. By commutativity, it follows that $a + 1 = 0$.

Let's find $a + a$:

- Look at the row for a . We have filled $a + 0 = a$ and $a + 1 = 0$. The only remaining element left from our field set $\{0, 1, a\}$ to complete this row is 1. Therefore, we are forced to conclude that $a + a = 1$.

+	0	1	a
0	0	1	a
1	1	a	0
a	a	0	1

Multiplication uses 1 as the multiplicative identity ($1 \cdot x = x$ for any $x \in \mathbf{F}$) and 0 as the zero element ($0 \cdot x = 0$). This fills every boundary spot immediately, leaving only one non-obvious entry to find: $a \cdot a$.

- By the multiplicative inverse axiom, every element except 0 must have an inverse that multiplies to 1. Since $1 \cdot 1 = 1$ and inverses are unique, 1 cannot be the inverse of a because $a \cdot 1 = a \neq 1$. Thus, a must be its own multiplicative inverse, forcing $a \cdot a = 1$.

·	0	1	a
0	0	0	0
1	0	1	a
a	0	a	1

Non-Textbook: 11. Explain why any two bases of the same finite-dimensional vector space must have the same number of elements.

Solution

Let V be a finite-dimensional vector space, and suppose B_1 and B_2 are both bases of V . To show they must have the same number of elements, we can apply the Span/LI Lemma (which states that the length of any linearly independent list of vectors cannot exceed the length of any spanning list of vectors).

By definition, a basis is simultaneously a linearly independent list and a spanning set for the vector space. We can just treat two bases with alternating roles as such:

- (a) Let B_1 be a linearly independent list in V , and let B_2 be a spanning set for V . By the Span/LI Lemma

$$\# \text{ of elements in } B_1 \leq \# \text{ of elements in } B_2$$

- (b) Conversely, let B_2 be a linearly independent list in V , and let B_1 be a spanning set for V . By the Span/LI Lemma

$$\# \text{ of elements in } B_2 \leq \# \text{ of elements in } B_1$$

Since $\# \text{ of elements in } B_1 \leq \# \text{ of elements in } B_2$ and $\# \text{ of elements in } B_2 \leq \# \text{ of elements in } B_1$ must both hold true, we are mathematically forced to conclude that:

$$\# \text{ of elements in } B_1 = \# \text{ of elements in } B_2$$

Thus, any two bases of the same finite-dimensional vector space must have the exact same number of elements.

Non-Textbook: 12. Let $\mathcal{P}_3(\mathbf{R})$ denote the vector space of real polynomials of degree at most 3. Define $T : \mathcal{P}_3(\mathbf{R}) \rightarrow \mathbf{R}^3$ by

$$T(p) = (p(0), p(1), p'(0)).$$

- (a) Prove that T is linear.
- (b) Compute $\ker T$.
- (c) Compute $\text{range } T$.
- (d) Verify the rank-nullity theorem for this example.

Solution

- (a) **Prove that T is linear.**

Let $p(x), q(x) \in \mathcal{P}_3(\mathbf{R})$ be any two arbitrary polynomials, and let $c \in \mathbf{R}$ be any arbitrary real scalar.

Additivity: We evaluate the transformation applied to the sum of the two polynomials, $(p+q)(x)$. By the definition of functions, we know that $(p+q)(x) = p(x) + q(x)$ and its derivative is $(p+q)'(x) = p'(x) + q'(x)$. Plugging this into the definition for T :

$$\begin{aligned} T(p+q) &= ((p+q)(0), (p+q)(1), (p+q)'(0)) \\ &= (p(0) + q(0), p(1) + q(1), p'(0) + q'(0)) \\ &= (p(0), p(1), p'(0)) + (q(0), q(1), q'(0)) \\ &= T(p) + T(q) \end{aligned}$$

Homogeneity: We evaluate the transformation applied to a scalar multiple of a polynomial, $(cp)(x)$. By definition, $(cp)(x) = c \cdot p(x)$ and its derivative is $(cp)'(x) = c \cdot p'(x)$. Plugging this into the definition for T :

$$\begin{aligned} T(cp) &= ((cp)(0), (cp)(1), (cp)'(0)) \\ &= (c \cdot p(0), c \cdot p(1), c \cdot p'(0)) \\ &= c \cdot (p(0), p(1), p'(0)) \\ &= cT(p) \end{aligned}$$

Since both additivity and homogeneity hold across all elements, T is a **linear map**.

(b) Compute $\ker T$.

Definition: The kernel/null space of a linear map consists of all input vectors that are mapped directly onto the zero element of the target space.

$$\text{Null } T = \{p \in \mathcal{P}_3(\mathbf{R}) : T(p) = (0, 0, 0)\}$$

Let us construct a general arbitrary polynomial $p \in \mathcal{P}_3(\mathbf{R})$:

$$p(x) = ax^3 + bx^2 + cx + d \quad \text{where } a, b, c, d \in \mathbf{R} \quad p'(x) = 3ax^2 + 2bx + c$$

We can now calculate the 3 components required by the linear map:

$$\begin{aligned} p(0) &= a(0)^3 + b(0)^2 + c(0) + d = d \\ p(1) &= a(1)^3 + b(1)^2 + c(1) + d = a + b + c + d \\ p'(0) &= 3a(0)^2 + 2b(0) + c = c \end{aligned}$$

which gives us

$$T(p) = (d, a + b + c + d, c)$$

and a system of equations:

$$\begin{aligned} (1) \quad & d = 0 \\ (2) \quad & a + b + c + d = 0 \\ (3) \quad & c = 0 \end{aligned}$$

We can substitute (1) and (3) into equation (2):

$$a + b + 0 + 0 = 0 \implies a + b = 0 \implies b = -a$$

And we get a as a free variable. Now we can substitute $b = -a$, $c = 0$, $d = 0$ back into $p(x)$:

$$p(x) = ax^3 + (-a)x^2 + 0x + 0 = a(x^3 - x^2)$$

Therefore, the kernel/null space consists of all scalar multiples of the polynomial $(x^3 - x^2)$:

$$\ker T = \{a(x^3 - x^2) \in \mathcal{P}_3(\mathbf{R}) : a \in \mathbf{R}\} = \text{span}(x^3 - x^2)$$

(c) Compute range T .

Definition: The range (or image) of a linear map is the collection of all output vectors that can be reached by giving the entire input space through the transformation T :

$$\text{range } T = \{T(p) \in \mathbf{R}^3 : p \in \mathcal{P}_3(\mathbf{R})\}$$

From part (b), we know that applying T to a general polynomial $p(x) = ax^3 + bx^2 + cx + d$ yields the output:

$$\begin{aligned} T(p) &= (d, a + b + c + d, c) \\ &= (0, a, 0) + (0, b, 0) + (0, c, c) + (d, d, 0) \\ &= a(0, 1, 0) + b(0, 1, 0) + c(0, 1, 1) + d(1, 1, 0) \end{aligned}$$

Then we can get:

$$\begin{aligned} \text{range } T &= \text{span}\{(0, 1, 0), (0, 1, 0), (0, 1, 1), (1, 1, 0)\} \\ &= \text{span}\{(1, 1, 0), (0, 1, 0), (0, 1, 1)\} \end{aligned}$$

Let us verify if these three vectors are linearly independent. If we arrange them as columns in a matrix to check their span:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

We can see that the columns are linearly independent (the matrix has full rank 3). Because we have found 3 linearly independent vectors inside the target vector space $\mathbf{R}^3$, and the total dimension of $\mathbf{R}^3$ is exactly 3, these vectors must span the entire target space.

$$\text{range } T = \mathbf{R}^3$$

(d) **Verify the rank-nullity theorem for this example.**

Theorem: The Rank-Nullity Theorem states that for any linear map $T : V \rightarrow W$ on a finite-dimensional vector space V :

$$\dim V = \dim(\text{Null } T) + \dim(\text{range } T)$$

(a) **Dimension of the Domain ($V = \mathcal{P}_3(\mathbf{R})$):** A standard basis for the space of polynomials of degree at most 3 is $\{1, x, x^2, x^3\}$.

$$\dim V = 4$$

(b) **Dimension of the Kernel ($\ker T$):** In part (b), we determined that $\ker T = \text{span}(x^3 - x^2)$.

$$\dim(\ker T) = 1$$

(c) **Dimension of the Range ($\text{range } T$):** In part (c), we determined that $\text{range } T = \mathbf{R}^3$.

$$\dim(\text{range } T) = 3$$

Substituting back into the theorem gives us:

$$\begin{aligned}\dim(\ker T) + \dim(\text{range } T) &= 1 + 3 \\ &= 4 \\ &= \dim V\end{aligned}$$

Non-Textbook: 13. This homework took me 5 hours of uninterrupted work, though I also review concepts and polish my notes while I finish my homework.